

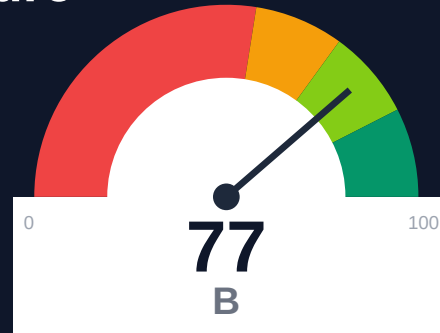
# Windmill

## Documentation Infrastructure Audit Report

Prepared by VeriOps

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Risk: Moderate

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### Key Metrics

Pages scanned: 979

Report type: Premium Executive Audit

Methodology: 5-pillar automated analysis + expert synthesis

### Headline Findings

- Broken internal links: 3
- Link verification inconclusive for 1 URLs (auth/rate-limit/server block); these are excluded
- SEO/GEO issue rate is high: 22.47%
- API usage-doc coverage is low: 66.67% (86 API pages detected)

This report is generated automatically by the VeriOps platform. Estimates are directional.

## Executive Summary



Documentation audit of windmill.dev/docs crawled 973 pages with 100% coverage against discovered scope. Site contains 3 confirmed broken internal links within user-facing documentation. API reference coverage stands at 66.67% across 86 detected API pages. SEO/geo issue rate is elevated at 22.43%, and only 53.65% of pages display last-updated metadata. Example code reliability estimate is 83.09% with overall audit confidence at 91.5%.

External posture: SEO/GEO issue rate **22.5%**, public API coverage **66.7%**, broken links **3**.

|                                         |                                      |                                            |
|-----------------------------------------|--------------------------------------|--------------------------------------------|
| <div>77</div> <div>Audit Score</div>    | <div>B</div> <div>Grade</div>        | <div>Moderate</div> <div>Risk Band</div>   |
| <div>979</div> <div>Pages Scanned</div> | <div>3</div> <div>Broken Links</div> | <div>22.5%</div> <div>SEO Issue Rate</div> |

### Per-Site Breakdown

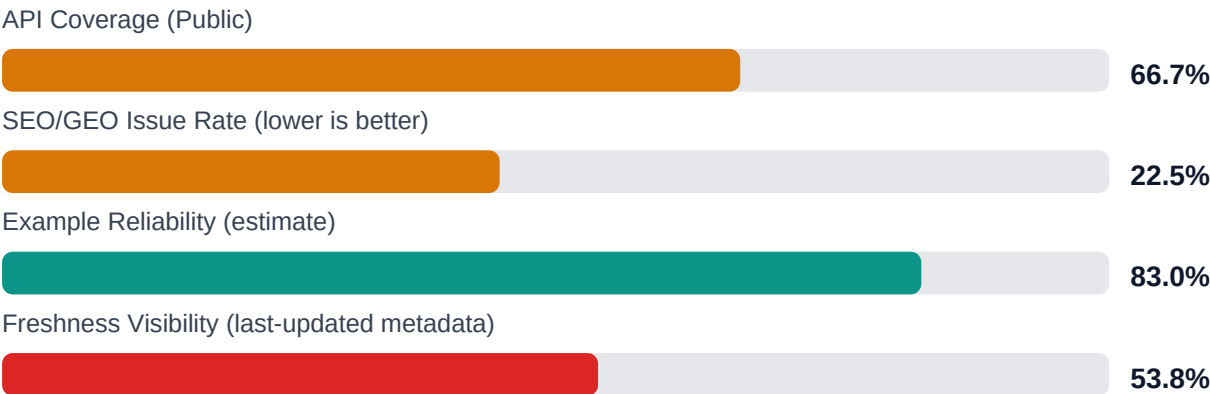
| Site URL                      | Pages | Broken Links | API Cov % | Example Rel % | SEO Issue % |
|-------------------------------|-------|--------------|-----------|---------------|-------------|
| https://www.windmill.dev/docs | 979   | 3            | 66.7      | 83.0          | 22.5        |

## Industry Benchmark Comparison

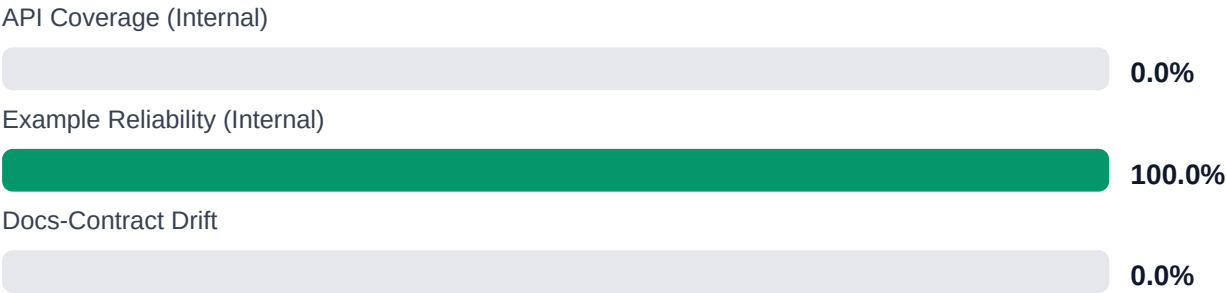
| Benchmark                         | Score | Gap |
|-----------------------------------|-------|-----|
| Top performers (Stripe, Twilio)   | 92+   | -15 |
| Industry average (developer docs) | 85    | -8  |
| Minimum acceptable                | 70    | +7  |
| Your score                        | 77    | -   |

Gap to industry average: **8 points**. Top-performing developer documentation platforms (Stripe, Twilio, Vercel) score 92+. Closing this gap reduces support ticket volume by an estimated 20-30%.

## Board-Level Metrics



### Internal Metrics (from scorecard)



## Financial Exposure Model

| Low Estimate                                                                                                                                               | Base Estimate                                                                                                                                               | High Estimate                                                                                                                                               |
|------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <div><b>\$140,822</b><br/>annual exposure</div> <div>Remediation: <b>\$2,090</b><br/>Monthly loss: <b>\$4,125</b><br/>Opportunity: <b>\$7,436/mo</b></div> | <div><b>\$249,513</b><br/>annual exposure</div> <div>Remediation: <b>\$5,445</b><br/>Monthly loss: <b>\$12,903</b><br/>Opportunity: <b>\$7,436/mo</b></div> | <div><b>\$431,728</b><br/>annual exposure</div> <div>Remediation: <b>\$8,800</b><br/>Monthly loss: <b>\$27,808</b><br/>Opportunity: <b>\$7,436/mo</b></div> |

Remediation: finding-level effort model | Monthly loss: operational friction model | Opportunity: engineering/support/release delay

Risk Matrix

| ID                     | Issue                                  | Severity | Monthly Loss | Fix Cost |
|------------------------|----------------------------------------|----------|--------------|----------|
| F-LAYERS               | Missing required doc layers            | HIGH     | \$1,782      | \$825    |
|                        |                                        |          |              |          |
| F-FRESHNESS            | Documentation freshness debt           | MEDIUM   | \$1,122      | \$550    |
|                        |                                        |          |              |          |
| F-RETRIEVAL            | RAG retrieval quality risk             | MEDIUM   | \$2,257      | \$990    |
|                        |                                        |          |              |          |
| F-API-COVERAGE         | Undocumented API operations            | LOW      | \$2,904      | \$1,100  |
|                        |                                        |          |              |          |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | LOW      | \$2,244      | \$935    |
|                        |                                        |          |              |          |
| F-DRIFT                | Code/docs contract drift               | LOW      | \$1,901      | \$660    |
|                        |                                        |          |              |          |
| F-TERMINOLOGY          | Terminology inconsistency              | LOW      | \$693        | \$385    |
|                        |                                        |          |              |          |

Priority Action Plan (Next 14 Days)

| ID                     | Finding                                | Severity |
|------------------------|----------------------------------------|----------|
| F-LAYERS               | Missing required doc layers            | HIGH     |
|                        |                                        |          |
| F-FRESHNESS            | Documentation freshness debt           | MEDIUM   |
|                        |                                        |          |
| F-RETRIEVAL            | RAG retrieval quality risk             | MEDIUM   |
|                        |                                        |          |
| F-API-COVERAGE         | Undocumented API operations            | LOW      |
|                        |                                        |          |
| F-EXAMPLES-RELIABILITY | Code examples fail or are non-runnable | LOW      |
|                        |                                        |          |

Recommended Actions

- Fix 3 confirmed broken internal documentation links immediately [impact: high, effort: low]
- Complete API reference documentation for the missing 33.33% of detected API surface (29 pages) [impact: critical, effort: high]

3. Add last-updated timestamps to 450+ pages missing freshness metadata [impact: medium, effort: medium]
4. Remediate SEO/geo issues affecting 218+ pages (22.43% of site) [impact: high, effort: medium]
5. Audit and fix example code reliability issues in the 16.91% flagged subset [impact: high, effort: medium]

## Expert Analysis: Strengths and Risks

### Strengths

- + Complete crawl coverage (100%) with 973 pages successfully examined
- + High overall audit confidence at 91.5% with strong link verification (95% confidence)
- + No repository navigation link breakage (repo\_broken\_links\_count: 0)
- + Example code reliability estimate above 80% (83.09%)
- + Single-product topology provides clear documentation structure

### Risks

- 3 confirmed broken internal links in user-facing documentation degrade user experience
- API reference coverage gap at 66.67% leaves one-third of API surface undocumented
- SEO/geo issue rate of 22.43% affects discoverability for 218+ pages
- Freshness metadata missing on 46.35% of pages (450+ pages) obscures content currency
- 16.91% of examples may contain reliability issues affecting developer onboarding
- 130 links excluded from verification may hide additional broken references
- API detection confidence at 85% suggests possible undercounted API surface

### Limitations

- \* External audit cannot verify accuracy or completeness of technical content, only structural and linking integrity
- \* API coverage detection at 85% confidence may undercount actual API surface; manual verification recommended for complete inventory
- \* Example reliability is estimated through static analysis; runtime validation and environment-specific testing not performed
- \* 130 excluded links were not verified and may contain additional broken references outside audit scope
- \* Freshness metadata absence does not confirm content is outdated, only that update tracking is not visible to users

## Methodology

This audit applies a rigorous, repeatable methodology combining automated analysis with expert synthesis. The platform evaluates documentation quality across five pillars.

### 1. Automated Crawl + Structural HTML Analysis

Every public documentation page is crawled and analyzed for structural integrity, link health, navigation depth, and content organization. The crawler detects broken links, orphaned pages, redirect chains, and missing metadata.

### 2. SEO/GEO Scoring Model (24 Checks)

Each page is evaluated against 8 GEO checks (LLM/AI search optimization) and 16 SEO checks (traditional search engine optimization). Checks cover meta descriptions, heading hierarchy, fact density, URL structure, internal linking, and content depth.

### 3. API Reference Cross-Coverage Analysis

The platform compares published API reference documentation against the actual API surface (OpenAPI specs, GraphQL schemas, gRPC protos). Coverage gaps, undocumented endpoints, and stale references are identified.

### 4. Code Example Reliability Estimation

Code examples embedded in documentation are analyzed for syntactic correctness, completeness, and consistency with current API signatures. Reliability scores estimate the likelihood of examples working as documented.

### 5. Executive Synthesis and Prioritization

All quantitative findings are synthesized into actionable executive narratives. The analysis identifies patterns, prioritizes risks, and generates strategic recommendations calibrated to business impact.

Appendix A -- Economic Model Assumptions

| Assumption                          | Value |
|-------------------------------------|-------|
| engineer_hourly_usd                 | 110.0 |
| support_hourly_usd                  | 50.0  |
| release_count_per_month             | 8.0   |
| baseline_manual_sync_hours_per_week | 12.0  |
| avg_release_delay_hours             | 4.5   |
| monthly_support_tickets             | 800.0 |
| docs_related_ticket_share           | 0.22  |
| avg_ticket_handling_minutes         | 24.0  |

Note: estimates are directional and should be calibrated with client rates, release cadence, and support volume.

Appendix B -- Evidence Traceability

| Finding                                                          | Evidence Source                                   | Confidence |
|------------------------------------------------------------------|---------------------------------------------------|------------|
| F-LAYERS -- Missing required doc layers                          | policy pack + frontmatter layer scan              | HIGH       |
| F-FRESHNESS -- Documentation freshness debt                      | frontmatter date scan                             | HIGH       |
| F-RETRIEVAL -- RAG retrieval quality risk                        | retrieval_evals_report.json                       | HIGH       |
| F-API-COVERAGE -- Undocumented API operations                    | OpenAPI + docs text scan                          | HIGH       |
| F-EXAMPLES-RELIABILITY -- Code examples fail or are non-runnable | examples_smoke_report.json                        | HIGH       |
| F-DRIFT -- Code/docs contract drift                              | pr_docs_contract.json + api_sdk_drift_report.json | HIGH       |
| F-TERMINOLOGY -- Terminology inconsistency                       | glossary.yml forbidden terms scan                 | HIGH       |

All findings are traceable to automated checks. Evidence sources reference specific crawl data, API specs, or code analysis results.